

FULL NAME: _____

BLOCK: _____

Rational Expressions — Equivalent & Simplify

Lesson 4-3 · Period 1 · Teacher Edition

OBJECTIVES

Math Objective	Write equivalent rational expressions over a restricted domain, identify domain restrictions from factored denominators, and simplify rational expressions by canceling common factors.
Language Objective	Explain, in writing and orally using sentence frames, why a value is excluded from the domain and why we cancel factors but not terms.
Essential Understanding	A rational expression equals an equivalent simpler form ONLY when both share the same domain — canceling a factor never erases the restriction it imposed.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Write equivalent rational expressions, simplify, and identify domain restrictions from polynomial denominators.
Essential Question	Why does canceling a common factor not remove its domain restriction?
Materials	Student packet, pencil, highlighter. Teacher displays worked Example 1 then gradual-release into Example 2.

[PIN] PRE-FLIGHT NOTE (read before class)

Explore is 7 bank items: Try-It 1 → #20 → #22 → #24 → Try-It 2 → #17 → #19 (extension). Paced for ~4–5 min per item.

If pairs finish early, push them to the Optional Reinforcement: Model & Discuss Parts B + C (verbal/TPS), then Practice #25.

If pace slows, cut #19 (extension) — it's the DOK-2 stretch; the core DOK-1 arc (Try-It 1 → #20/#22/#24 → Try-It 2 → #17) is the minimum viable set.

Tag: notice & wonder

Do Now

DOK: 2

Minutes: 5

Teacher does

- Project the graph (or use printed Do Now sheet).
- Circulate silently. Do NOT teach during this phase.
- Collect at 5 min sharp.

Students do

- Write 1 notice + 1 wonder.
- Write one sentence about what the equation must be doing at the skipped x -value.

Questions to ask

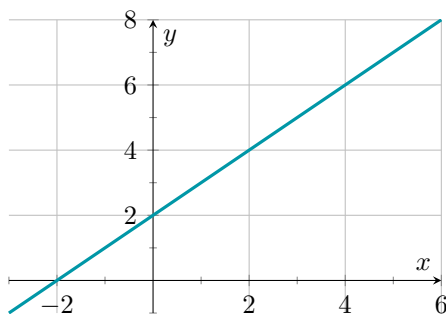
- (None during the phase — teacher harvests answers in Launch.)

Adult role

Solo teacher: circulate silently. Resist the urge to give hints.

Do Now (4-3-savvas-model-discuss-lesson-4-3-launch)**EXPLORE & REASON**

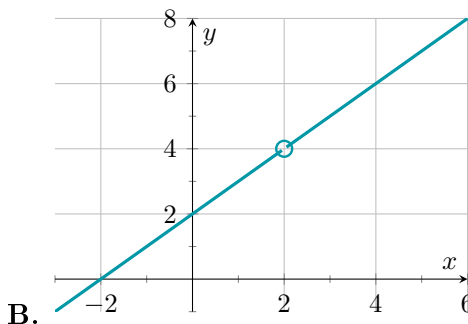
Consider the following graph of the function $y = x + 2$.



- A. What is the domain of this function?
- B. Sketch a function that resembles the graph, but restrict its domain to exclude 2.
- C. Use Structure Consider the function you have sketched. What kind of function might have a graph like this? Explain.

Sample Student Work

- A. all real numbers.



- C. An equation where the denominator equals zero when $x = 2$. For example, $y = (2x - 4)/(x - 2)$.

[CHECK] ANTICIPATED STUDENT RESPONSES

NOTICE (typical): “There’s a hole in the graph.” “The line has a gap.” “One point is missing.”

WONDER (typical): “Why is there a hole?” “What happens at that x ?” “Can you have an equation that does this?”

SENTENCES: students might write “the denominator equals zero at that x ” or “the equation is undefined there.”

KEY MOVE in Launch: surface the insight that when the denominator = 0, the expression is undefined — this is a domain restriction.

Tag: teacher models Example 1 + Example 2

Launch**DOK: 1–2****Minutes: 12****Teacher does**

- Bridge from Do Now: “You noticed the graph skips a point. That happens because the denominator equals zero at that x -value — we call it a domain restriction.”
- Project Example 1 (Savvas Topic 4-3). Think aloud: factor numerator and denominator, identify what x -values make the denominator zero, write the equivalent expression with restriction stated.
- Project Example 2. Think aloud: factor completely, cancel the common factor, but keep the domain restriction from the canceled factor in the final answer.
- Pause 1× for 30-sec partner-check: “What x -value is excluded in Example 1?”
- Do NOT solve Try-Its. Release promptly at minute 17.

Students do

- Watch silently through Example 1.
- During partner-check: identify the excluded x -value.
- Copy the Equivalent + Simplify Rules card into their page margin.
- Note the COMMON ERROR warning: cancel only factors, never terms.

Questions to ask

- “What values make the denominator zero?”
- “Can I cancel this factor? Is it a factor of the entire numerator?”
- “After canceling, does the domain restriction disappear?”

Adult role

Project, think aloud. Do NOT give students Try-Its to work until minute 17.

[MIC] TEACHER SCRIPT

1. “From the Do Now, you noticed the graph skips a point. That x -value is excluded from the domain because the denominator equals zero there.”
2. Example 1 walk-through: “Watch me. I factor top and bottom. Any x that makes the denominator zero is restricted — I write $x \neq$ that value next to my answer.”
3. Example 2 walk-through: “Now I simplify. I factor, find the matching factor in numerator and denominator, and cancel it. But — critical — the restriction from that canceled factor STAYS. Canceling doesn’t erase the hole.”
4. Common-error flag: “Don’t cancel terms that are added or subtracted. Factor FIRST, then cancel only matching factor pairs.”
5. Release: “Now you try. Try-It 1 asks for an equivalent expression. Try-It 2 asks you to simplify completely.”

Tag: Think-Pair-Share + Practice

Explore**DOK: 1–2****Minutes: 33****Teacher does**

- 5 laps circulation across 33 min (~6–7 min each).
- Lap 1: Try-It 1 — press pairs to factor before writing the equivalent form.
- Lap 2: #20 + #22 — check that pairs identify domain restrictions correctly.
- Lap 3: #24 + Try-It 2 — press pairs to factor completely before canceling.
- Lap 4: #17 — verify pairs state the domain restriction alongside the simplified form.
- Lap 5: #19 (extension) — stretch. Cut if pace slows.
- Do not give answers. Use sentence frame to press justification.

Students do

- Pairs move in order through 7 items: 2 Try-Its + 5 Practice.
- For each item: factor completely FIRST, then cancel, then state restriction.
- For Practice #19 (simplify with domain restrictions): show all steps.
- Justify with sentence frame before writing.

Questions to ask

- Did you factor completely before canceling?
- What x -values make the denominator zero?
- After simplifying, what is the domain restriction?
- Can you use the sentence frame to explain your answer?
- For #19: what is the domain restriction from the canceled factor?

Adult role

Solo teacher: 5 laps. Press pairs to justify using the rule card. No answers.

Try It 1 (4-3-savvas-try-it-1-lesson-4-3)

Write an expression equivalent to $(x - 4)/(x)$ over the domain $\{x \mid x \neq 0 \text{ or } -2\}$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $(x^2 - 2x - 8)/(x^2 + 2x)$

Practice #20 — Identify equivalent (4-3-savvas-q20)

Write an equivalent expression over the given domain.

$$\frac{x(x - 2)}{x - 5} \text{ for all } x \text{ except } 5 \text{ and } -6$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $(x^3 + 4x^2 - 12x)/(x^2 + x - 30)$

Practice #22 (4-3-savvas-q22)

What is the simplified form of the rational expression? What is the domain?

$$\frac{y^2 - 5y - 24}{y^2 + 3y}$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $(y - 8)/(y)$ for all y except 0 and -3 .

Practice #24 (4-3-savvas-q24)

What is the simplified form of the rational expression? What is the domain?

$$\frac{x^2 + 8x + 15}{x^2 - x - 12}$$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $(x + 5)/(x - 4)$, $x \neq -3, 4$.

Try It 2 (4-3-savvas-try-it-2-lesson-4-3)

Simplify each expression and state the domain.

a. $\frac{x^2 + 2x + 1}{x^3 - 2x^2 - 3x}$

b. $\frac{x^3 + 4x^2 - x - 4}{x^2 + 3x - 4}$

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): a. $(x + 1)/(x(x - 3))$ for all x except 0, -1 , and 3. b. $x + 1$ for all x except 1 and -4 .

Practice #17 — Simplify (4-3-savvas-q17)

Construct Arguments Explain how you can tell whether a rational expression is in simplest form.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): A rational expression is in simplest form when the numerator and denominator have no common factors other than 1.

Practice #19 — Simplify with domain restrictions (4-3-savvas-q19)

Reason If the denominator of a rational expression is $x^3 + 3x^2 - 10x$, what value(s) must be restricted from the domain for x ?

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $-5, 0, 2$

Teacher does

- Cold-call 2 pairs to read their sentence-frame sentence.
- Write one student-generated sentence on board.
- Preview P2: “Next class we’ll multiply and divide rational expressions — the same factor-and-cancel logic applies.”

Students do

- Presenting pair reads their sentence.
- Class signals thumbs up/sideways.
- Note: next class extends this to multiplication and division.

Questions to ask

- “What’s a sentence that captures today’s rule about domain restrictions?”
- “Why can’t we just ignore the restriction after we cancel?”

Adult role

Cold-call 2 pairs. Keep it brief — 5 min.

Tag: summary recap (not graded)

Exit Ticket**DOK: 1-2****Minutes: 2****Teacher does**

- Collect at door.
- Scan the fill-in items — misconceptions about domain restrictions tell you what to re-hit in P2 Do Now.

Students do

- Complete the fill-in stems.
- Be honest about confusion.

Questions to ask

- Did students correctly identify domain restrictions from the original denominator?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

To simplify $(x^2 + 3x - 10)/(x^2 - 4)$ I first factor to ____.

The restricted domain values are $x \neq$ ____.

After canceling the factor ____, my simplified expression is ____.

One biggest thing I learned today: _____

[CLOCK] PERIOD A & F — 65-MIN CLASS NOTES

Both sections get 65 min for P1. Use the extra 10 min on Explore, NOT Launch.

Suggested add: project Savvas Practice #22 and #24 on screen for 2 more TPS rounds if pairs finish fast.

Or: extend Model & Discuss Parts B + C with “write one more rational expression that has the same restriction — then simplify it.”

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Equivalent + Simplify Rules card as a sticker/cut-out on their desk.
- For Try-It 1, allow students to factor with a reference list of factoring patterns.
- Accept verbal sentence-frame completions in lieu of written.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames printed on student packet (don’t skip).
- Word card: “domain restriction” = an x -value that makes the denominator zero; excluded from the domain.
- “cancel” = divide out a common factor from numerator and denominator.
- “factor” = rewrite as a product of simpler expressions.
- Pair ELL students with English-dominant partners for TPS.